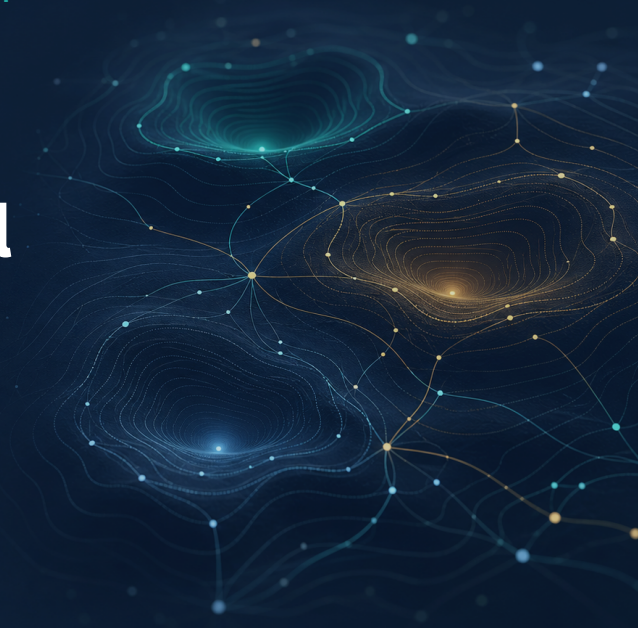
PRELIMINARY RESULTS · SCIENTIFIC ADVISORY UPDATE

Resting-state circuit profiles are operational

FOR2107 adaptation of Tozzi et al. (2024)

75 scans · 67 selected · 3 rest-FC features

23 July 2026 | Decision focus: what is demonstrated—and what is not



Tozzi in one frame

Rest + tasks
D/S/A · faces · Go/NoGo

→

41 regional
circuit scores

→

Healthy norm
participant-level SD

→

Geometry
$d_{ij} = 1 - r_{ij}$

→

Average linkage
candidate biotypes

→

Convergence
stability · nulls · clinical

Scientific objective

Replace diagnostic averages with interpretable, individual circuit-dysfunction profiles—and test whether profile-defined groups differ clinically.

Landscape intuition



valleys ≈ similar circuit profiles

Important distinction

Tozzi does **not** fit an explicit energy function. “Landscape” is only an intuition for basins in correlation-distance feature space.

Task conditions contribute condition-specific activation/connectivity scores; task-free connectivity is quantified independently.

Source: Tozzi et al., *Nature Medicine* 2024 [1]. Diagram recreated; no publication figure copied.

Current implementation status

Implemented

- 75 external MNI-2 mm scans ingested
- Tozzi ROI extraction; Fisher-*z* FC
- Healthy-reference standardization
- $1 - r$ distance + average linkage
- Null/permutation, LOO, 80%, split-half
- HTML/PDF report + 16 passing tests

Partial

- Preprocessing accepted, not reproduced
- Provenance incomplete for cleaned scans
- Motion QC unavailable; no FD exclusion
- MDD/anxiety phenotype mapping operational
- Validation runs = 50, not 10,000
- Single site; small clinical sample

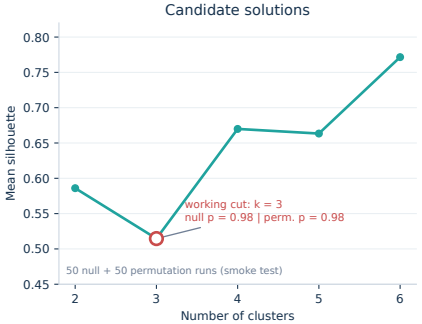
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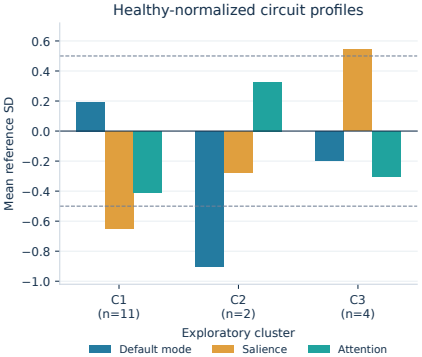
- Face-matching task contrasts + FC
- Affective-priming contrasts + FC
- Go/NoGo equivalent in FOR2107
- Independent/leave-study-out validation
- Treatment-response replication
- Predeclared replication decision rule

Operational now: a rest-only methodological reproduction. **Not yet:** a multimodal replication or validated biotype result.

Preliminary scientific results

75	67	17	50	3
REST SCANS	SELECTED	CLINICAL	REFERENCE	FC FEATURES





Method-level match

Same ROI-circuit summaries, healthy normalization, correlation distance, and average linkage for the available rest modality.

Observation

Circuit-profile separation is visible; working $k = 3$ sizes are 11/2/4. Split-half mean profile correlation = 0.74.

Scientific boundary

$k = 3$ is not above either null ($p = .98/.98$); no symptom effect survives FDR; motion is unassessed for all 75.

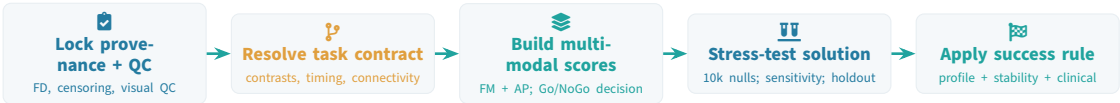
Comparison to Tozzi

	Tozzi et al.	Current work	
Data	801 clinical + 137 controls; multiple studies	17 clinical + 50 controls selected; one site; 75 scans total	!
Inputs	41 task-free + task-evoked circuit scores	3 within-circuit resting-state FC scores	!
Task probes	Rest; conscious/nonconscious faces; Go/NoGo	Rest only; task mappings pending; no Go/NoGo equivalent	✗
State representation	Individual HC-referenced circuit-score vector	Same representation for available rest circuits	✓
"Landscape"	No explicit energy model; $1 - r$ profile geometry	Same geometry; landscape used only as metaphor	✓
Clustering	Correlation distance; average linkage; six viable biotypes	Same algorithm; working $k = 3$ only for pipeline exercise	✓
Validation	10,000× null/permutation; LSO/20%; split-half	50× smoke tests; LOO/80%; split-half; no holdout	!
External effects	Symptoms, cognition, treatment response	Four symptom totals; no FDR-significant result; no treatment	✗
Inference	Clinically distinct multimodal biotypes	**No biotype evidence**; exploratory rest-only profiles	−

✓ matching implementation ! partial/underpowered ✗ unavailable − impossible to assess now

Sources: Tozzi et al. [1]; FOR2107 context [2].

Next scientific milestones



Replication criteria—define before scaling

- Same signed circuit-profile structure, not same labels
- Stability above predeclared ARI/profile thresholds
- Better-than-null clustering under full iteration counts
- Robust to motion, preprocessing, *k*, and feature choices
- Replicable external symptom/behavior associations

Critical design decision

FOR2107 has no direct Go/NoGo equivalent. Predeclare whether the target is:

partial replication
rest + affect only

or

construct-aligned extension
memory/control proxy

Current assessment: technically operational · scientifically informative as a feasibility result · not yet a replication claim.